

Class Test I August 2023
2nd Year Electronics & Telecomm Engg. (Section A & B)
3ETRC3 Data Structure

Time: 70 min

Max Marks: 20

Note: 1. Attempt any four questions. All questions carry equal marks.

2. Write the output with sample input while answering programming question

Q.1 Write C program to add two integers using functions and pointers.

Q.2 Define Data Structures. What are different types of data structures?

Q.3 What all operations are possible on Data Structures? Explain in brief.

Q.4 Write a C program for inserting an element in 1D array other than last position.

Q.5 For 2D array declared as: `int arr[][2]={1,2,3,4};`

- i. draw the memory map for row major format.
- ii. find out size of array in Bytes.
- iii. find out number of rows.
- iv. find out number of elements in array.
- v. for base address = 1000, find out address of `arr[1][1]`.